Vidyalankar Polytechnic	DEPARTMENT OF COMPUTER ENGINEERING

Subject: - Mobile Application Development	**Subject Code**: 316006
Semester: - VI	**Course**: CO6K-B
Laboratory No: V119	**Name of Subject Teacher**: Prasad Koyande
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Experiment No:	9
Title of Experiment	Write program to implement Linear layout and Constraint layout to create any registration form with Custom Toast Alert.

Layout and UI Design

1. Design a registration form using LinearLayout that includes fields for Name, Email, Phone, and a Register button.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com
/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name" />

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email"

android:inputType="textEmailAddress" />

    <EditText
```

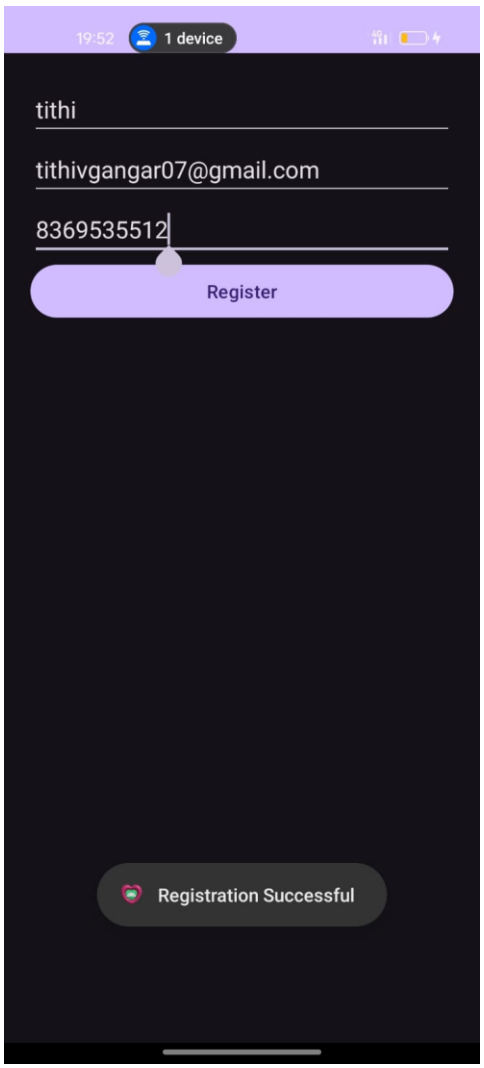
MainActivity.java

```
package com.example.tithiexp9;

import android.os.Bundle;
import android.widget.*;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    EditText etName, etEmail, etPhone;
    Button btnRegister;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        etEmail = findViewById(R.id.etEmail);
        etPhone = findViewById(R.id.etPhone);
        btnRegister =
findViewById(R.id.btnRegister);
```

```
        android:id="@+id/etPhone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone"
        android:inputType="phone" />

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register" />

</LinearLayout>
```

```
        btnRegister.setOnClickListener(v -> {
            Toast.makeText(this, "Registration
Successful", Toast.LENGTH_SHORT).show();
        });
    }
}
```

Output



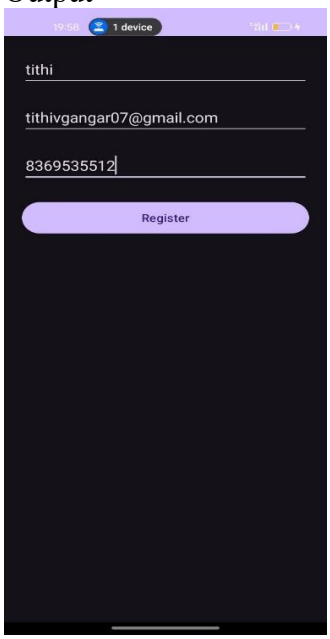
2. Recreate the same form using ConstraintLayout, ensuring proper alignment and spacing between components.

activity_main.xml	MainActivity.java

<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
 	
xmlns:android="http://schemas.android.com/apk/res/android"
 	
xmlns:app="http://schemas.android.com/apk/res-auto"
android:layout_width="match_parent"
android:layout_height="match_parent"
android:padding="20dp">
 	
<EditText
android:id="@+id/etName"
android:layout_width="0dp"
android:layout_height="wrap_content"
android:hint="Enter Name"
 	
app:layout_constraintTop_toTopOf="parent"
 	
app:layout_constraintStart_toStartOf="parent"
 	
app:layout_constraintEnd_toEndOf="parent"/>
 	
<EditText
android:id="@+id/etEmail"
android:layout_width="0dp"
android:layout_height="wrap_content"
android:hint="Enter Email"
android:inputType="textEmailAddress"
 	
app:layout_constraintTop_toBottomOf="@id/etName"
 	
app:layout_constraintStart_toStartOf="parent"
 	
app:layout_constraintEnd_toEndOf="parent"
android:layout_marginTop="16dp"/>
 	
<EditText
android:id="@+id/etPhone"
android:layout_width="0dp"
	package com.example.tithiexp9;
	
	import android.os.Bundle;
	import android.widget.Toast;
	import androidx.appcompat.app.AppCompatActivity;
	
	public class MainActivity extends AppCompatActivity {
	
	@Override
	protected void onCreate(Bundle savedInstanceState) {
	super.onCreate(savedInstanceState);
	
	setContentView(R.layout.*activity_main*);
	}
	}

```
        android:layout_height="wrap_content"
        android:hint="Enter Phone"
        android:inputType="phone"

app:layout_constraintTop_toBottomOf="@id/etEmail"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp"/>

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:text="Register"

app:layout_constraintTop_toBottomOf="@id/etPhone"

app:layout_constraintStart_toStartOf="parent"

app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="20dp"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

Output



3. Add gender selection using RadioGroup and display the selected gender in the custom toast.

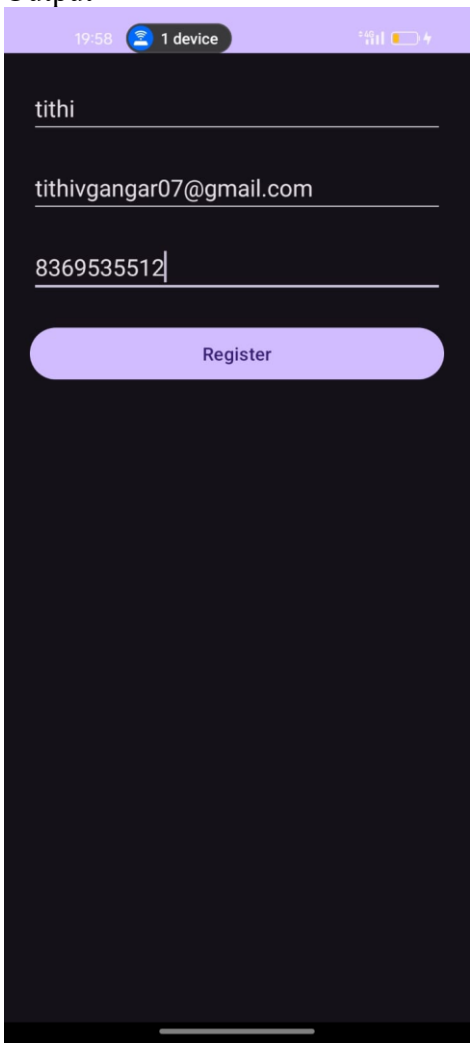
activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">

   <RadioGroup
      android:id="@+id/radioGroup"
      android:layout_width="wrap_content"

android:layout_height="wrap_content">

      <RadioButton
         android:id="@+id/rbMale"

android:layout_width="wrap_content"

android:layout_height="wrap_content"
         android:text="Male" />

      <RadioButton
         android:id="@+id/rbFemale"

android:layout_width="wrap_content"

android:layout_height="wrap_content"
         android:text="Female" />
   </RadioGroup>

   <Button
      android:id="@+id/btnSubmit"
      android:layout_width="match_parent"

android:layout_height="wrap_content"
      android:text="Submit" />

</LinearLayout>
```

MainActivity.java

```
package com.example.tithiexp9;

import android.os.Bundle;
import android.widget.*;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

   RadioGroup radioGroup;
   Button btnSubmit;

   @Override
   protected void onCreate(Bundle
savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      radioGroup =
findViewById(R.id.radioGroup);
      btnSubmit =
findViewById(R.id.btnSubmit);

      btnSubmit.setOnClickListener(v -> {

         int selectedId =
radioGroup.getCheckedRadioButtonId();

         if (selectedId == -1) {
            Toast.makeText(this, "Please select
gender", Toast.LENGTH_SHORT).show();
         } else {
            RadioButton selected =
findViewById(selectedId);
            Toast.makeText(this, "Selected: " +
selected.getText(),

Toast.LENGTH_SHORT).show();
         }
      });
   }
}
```

Output



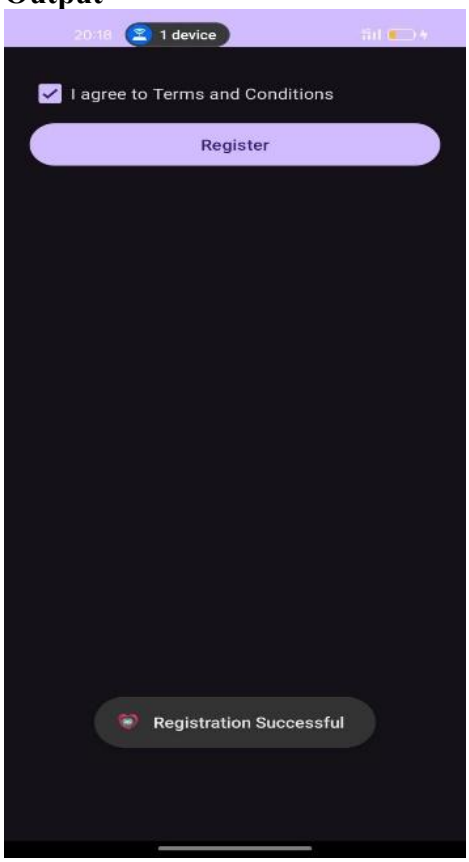
4. Include a CheckBox for "Terms and Conditions" and validate that it is checked before showing the toast.

activity_main.xml	**MainActivity.java**
<?xml version="1.0" encoding="utf-8"?> <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" android:layout_width="match_parent" android:layout_height="match_parent" android:orientation="vertical" android:padding="20dp"> <CheckBox android:id="@+id/cbTerms" android:layout_width="wrap_content"	package com.example.tithiexp9; import android.os.Bundle; import android.widget.*; import androidx.appcompat.app.AppCompatActivity ; public class MainActivity extends AppCompatActivity { CheckBox cbTerms; Button btnRegister; @Override protected void onCreate(Bundle

```
        android:layout_height="wrap_content"
        android:text="I agree to Terms and
Conditions" />

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register" />

</LinearLayout>
```

```
savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        cbTerms =
findViewById(R.id.cbTerms);
        btnRegister =
findViewById(R.id.btnRegister);
        btnRegister.setOnClickListener(v -> {
            if (cbTerms.isChecked()) {
                Toast.makeText(this, "Registration
Successful",

Toast.LENGTH_SHORT).show();
            } else {
                Toast.makeText(this, "Please accept
Terms & Conditions",

Toast.LENGTH_SHORT).show();
            }
        });
    }
}
```

Output



Input Validation and Feedback

1 Implement input validation to ensure all fields are filled before allowing registration.

activity_main.xml

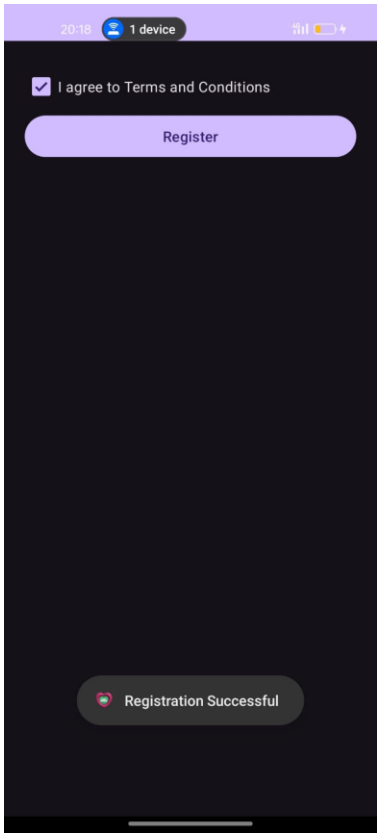
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <EditText
        android:id="@+id/etName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Name" />

    <EditText
        android:id="@+id/etEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Email" />

    <EditText
        android:id="@+id/etPhone"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Phone" />

    <Button
        android:id="@+id/btnRegister"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Register" />

</LinearLayout>
```

MainActivity.java

```
package com.example.tithiexp9;

import android.os.Bundle;
import android.widget.*;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

    EditText etName, etEmail, etPhone;
    Button btnRegister;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {
        super.onCreate(savedInstanceState);

setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        etEmail = findViewById(R.id.etEmail);
        etPhone =
findViewById(R.id.etPhone);
        btnRegister =
findViewById(R.id.btnRegister);

        btnRegister.setOnClickListener(v -> {

            String name =
etName.getText().toString().trim();
            String email =
etEmail.getText().toString().trim();
            String phone =
etPhone.getText().toString().trim();

            if (name.isEmpty() || email.isEmpty()
|| phone.isEmpty()) {
                Toast.makeText(this, "All fields
are required!",

Toast.LENGTH_SHORT).show();
```

	} else {
	Toast.*makeText*(this, "Registration Successful",
	
	Toast.*LENGTH_SHORT*).show();
	}
	});
	}
	}

Output



2 Display a custom toast with the user's name and email after successful registration.

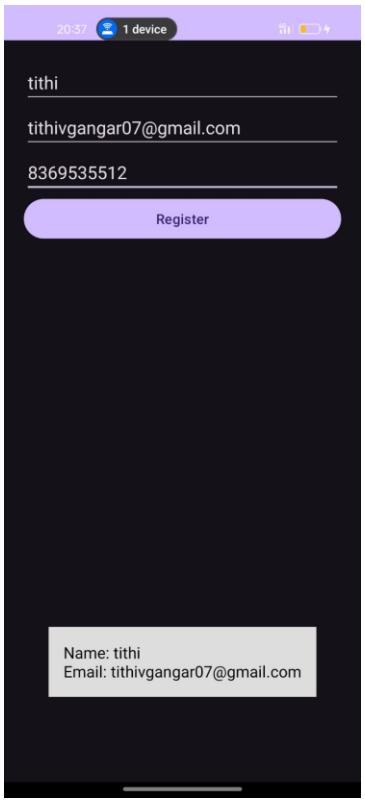
toast_layout.xml	**MainActivity.java**
<?xml version="1.0" encoding="utf-8"?> <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" android:layout_width="wrap_content" android:layout_height="wrap_content" android:orientation="vertical" android:padding="15dp"	package com.example.tithiexp9; import android.os.Bundle; import android.view.*; import android.widget.*; import androidx.appcompat.app.AppCompatActivity;

```
    android:background="#DDDDDD">

    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textColor="#000"
        android:textSize="16sp" />

</LinearLayout>
```

```
public class MainActivity extends
AppCompatActivity {

    EditText etName, etEmail;
    Button btnRegister;

    @Override
    protected void onCreate(Bundle
savedInstanceState) {

super.onCreate(savedInstanceState);

setContentView(R.layout.activity_mai
n);

        etName =
findViewById(R.id.etName);
        etEmail =
findViewById(R.id.etEmail);
        btnRegister =
findViewById(R.id.btnRegister);

        btnRegister.setOnClickListener(v
-> {

            String name =
etName.getText().toString();
            String email =
etEmail.getText().toString();

            LayoutInflater inflater =
getLayoutInflater();
            View layout =
inflater.inflate(R.layout.toast_layout,
null);

            TextView tvMessage =
layout.findViewById(R.id.tvMessage);
            tvMessage.setText("Name: " +
name + "\nEmail: " + email);

            Toast toast = new
Toast(getApplicationContext());

toast.setDuration(Toast.LENGTH_LO
NG);
            toast.setView(layout);
```

	toast.show();
	});
	}
	}

Output



3 Modify the toast layout to include an icon and change background color based on success or error.

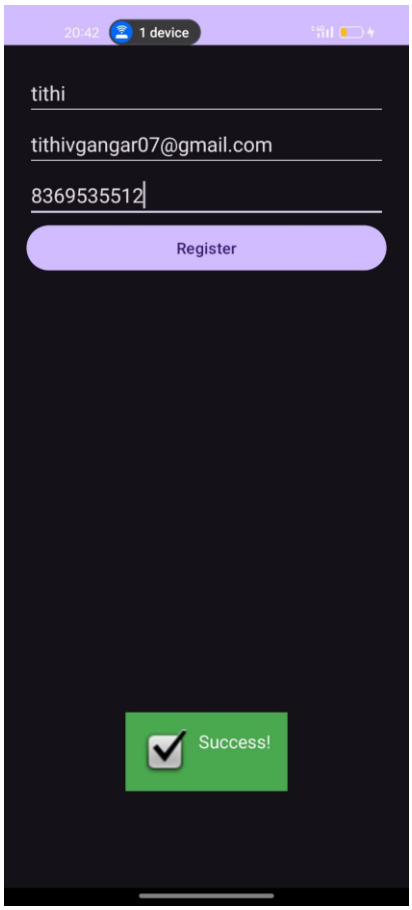
toast_layout.xml	**MainActivity.java**
<?xml version="1.0" encoding="utf-8"?> <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android" android:id="@+id/toastLayout" android:layout_width="wrap_content" android:layout_height="wrap_content" android:orientation="horizontal" android:padding="15dp"> <ImageView	package com.example.tithiexp9; import android.os.Bundle; import android.view.*; import android.widget.*; import androidx.appcompat.app.AppCompatActivity; public class MainActivity extends AppCompatActivity { EditText etName; Button btnRegister;

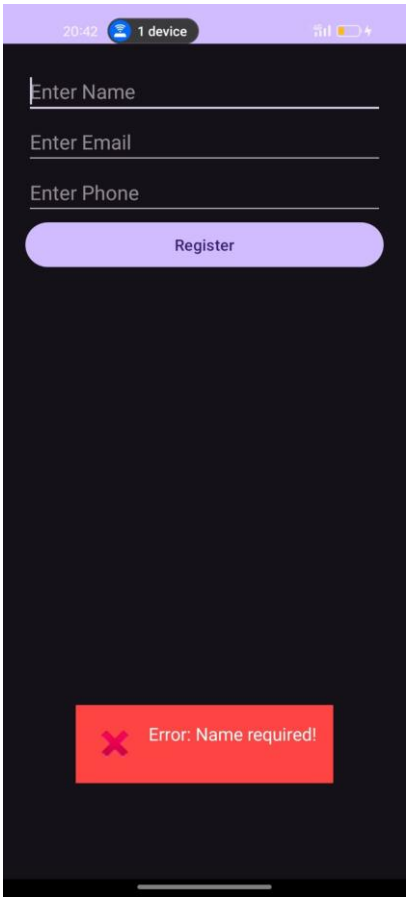
```
        android:id="@+id/imgIcon"
        android:layout_width="40dp"
        android:layout_height="40dp"
        android:layout_marginEnd="10dp"

android:src="@android:drawable/ic_dialog_info" />

    <TextView
        android:id="@+id/tvMessage"

android:layout_width="wrap_content"

android:layout_height="wrap_content"
        android:textColor="#FFFFFF"
        android:textSize="16sp" />

</LinearLayout>
```

```
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        btnRegister = findViewById(R.id.btnRegister);

        btnRegister.setOnClickListener(v -> {

            LayoutInflater inflater = getLayoutInflater();
            View layout = inflater.inflate(R.layout.toast_layout, null);

            LinearLayout toastLayout = layout.findViewById(R.id.toastLayout);
            ImageView imgIcon = layout.findViewById(R.id.imgIcon);
            TextView tvMessage = layout.findViewById(R.id.tvMessage);

            if (etName.getText().toString().isEmpty()) {


toastLayout.setBackgroundColor(0xFFFF4444); // Red

imgIcon.setImageResource(android.R.drawable.ic_delete);
                tvMessage.setText("Error: Name required!");

            } else {


toastLayout.setBackgroundColor(0xFF4CAF50); // Green

imgIcon.setImageResource(android.R.drawable.checkbox_on_background);
                tvMessage.setText("Success!");

            }
```

```
            Toast toast = new
Toast(getApplicationContext());

toast.setDuration(Toast.LENGTH_SHORT);
            toast.setView(layout);
            toast.show();
        });
    }
}
```

Output





4 Create a toast that appears only if the phone number is exactly 10 digits.

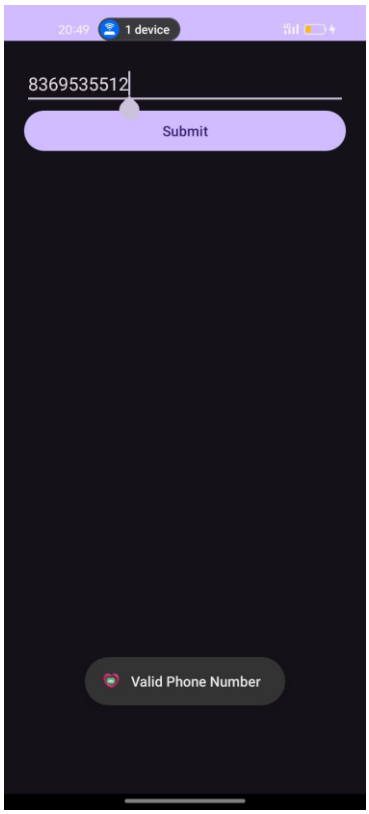
activity_main.xml

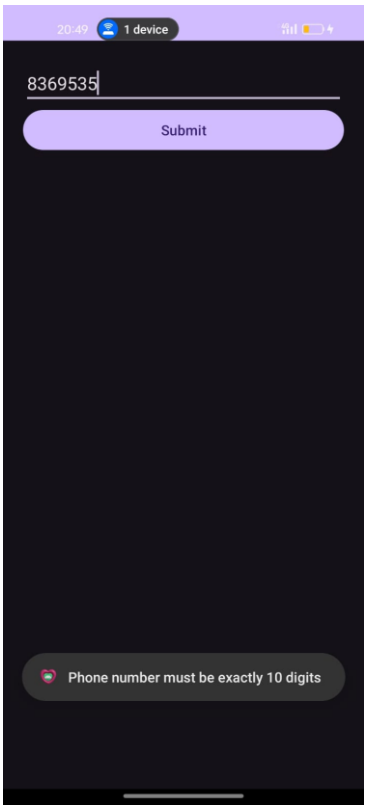
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
xmlns:android="http://schemas.android.com
/apk/res/android"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">

   <EditText
      android:id="@+id/etPhone"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:hint="Enter Phone"
      android:inputType="phone"
      android:maxLength="10"/>

   <Button
      android:id="@+id/btnSubmit"
      android:layout_width="match_parent"
      android:layout_height="wrap_content"
      android:text="Submit" />

</LinearLayout>
```

MainActivity.java

```
package com.example.tithiexp9;

import android.os.Bundle;
import android.widget.*;
import
androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends
AppCompatActivity {

   EditText etPhone;
   Button btnSubmit;

   @Override
   protected void onCreate(Bundle
savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

      etPhone = findViewById(R.id.etPhone);
      btnSubmit =
findViewById(R.id.btnSubmit);

      btnSubmit.setOnClickListener(v -> {

         String phone =
etPhone.getText().toString();

         if (phone.length() == 10) {
            Toast.makeText(this, "Valid Phone
Number",

Toast.LENGTH_SHORT).show();
         } else {
            Toast.makeText(this, "Phone
number must be exactly 10 digits",

Toast.LENGTH_SHORT).show();
         }
      });
   }
}
```

Output:





Marks Obtained			Dated signature of Teacher
Process Related (15)	Product Related (10)	Total (25)	